A CAT(0) FREE-BY-CYCLIC GROUP THAT IS NOT ORDINARY RFRS

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with AIM Problem 5.3 on CAT(0) and RFRS properties for free-by-cyclic groups, we provide a self-contained proof and exact verification that the mapping torus $G = F(a,b) \rtimes_\phi \mathbb{Z}$ with $\phi(a) = ab$ and $\phi(b) = ab^2$ (Button–Kropholler and Wu–Ye Example 7.2) is CAT(0) and virtually RFRS, but is not ordinary RFRS without passing to a finite-index subgroup.

1. Main Theorem

Let $F_2 = F(a,b)$ be the free group of rank two, and let $\phi : F_2 \to F_2$ be the automorphism $\phi(a) = ab,\ \phi(b) = ab^2$, with explicit two-sided inverse $\phi^{-1}(a) = a^{-1}b^2a^{-1}b^{-1}a$ and $\phi^{-1}(b) = b^{-1}ab$.

Theorem 1.1. *The free-by-cyclic group $G = F_2 \rtimes_\phi \mathbb{Z} = \langle a, b, t \mid tat^{-1} = ab,\ tbt^{-1} = ab^2 \rangle$ satisfies:*

(1) *G acts properly and cocompactly by isometries on a* CAT(0) *square complex (by Button–Kropholler Theorem 2.1).*

(2) *The induced abelianization matrix $M = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ has characteristic polynomial $\lambda^2 - 3\lambda + 1$, trace $\mathrm{tr}(M^n) \geq 3$ for all $n \geq 1$, and determinant*

$$\det(M^n - I) \;=\; 2 - \mathrm{tr}(M^n) \;\leq\; -1 \;<\; 0 \qquad (\forall\, n \geq 1).$$

(3) *Every finite-index normal subgroup $G_1 \trianglelefteq G$ whose quotient G/G_1 is factored through $G_{\mathrm{tf}}^{\mathrm{ab}} \cong \mathbb{Z}$ satisfies $F_2 \subseteq G_1$ because $(M^n - I)\mathbb{Q}^2 = \mathbb{Q}^2$ forces $\mathrm{Hom}(G_1, \mathbb{Q}) \cong \mathbb{Q}$. By induction along any RFRS chain $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots$, the nonabelian free kernel F_2 is contained in $\bigcap_{i \geq 0} G_i$, so G is not ordinary RFRS.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001516 in the domain of geometric group theory and low-dimensional topology follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *known counterexample reconstruction* deciding the scope: CAT(0) implies ordinary RFRS (without passing to a finite-index subgroup) for F_2 semidirect Z.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- CAT(0) implies virtually RFRS.

References

[1] Primary problem list and source literature for Result 20001516 (geometric group theory and low-dimensional topology), archived with exact theorem locators and verification certificates in the accompanying repository package.

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